

Hamroun Sami Abdelmalek

Full-Stack Web Developer (Backend-Focused)

hamrounsamiabdelmalek@gmail.com | +213 559 40 99 13 | El Mohammadia, Algiers | Ready-to-work

PROFESSIONAL SUMMARY

Backend-leaning full-stack developer and 3rd-year Computer Engineering student (Cybersecurity, USTHB) with a proven track record of shipping production-grade web applications. Built and deployed 6+ projects across Node.js/Express, Flask, React, and PostgreSQL — including a multi-role agricultural marketplace with real-time GPS tracking and an automated exam supervision system used at faculty level. Competed in CTF web exploitation events and participates in hackathons. Available immediately for remote or hybrid roles.

TECHNICAL SKILLS

Backend	Node.js · Express.js · Flask (Python) · REST API Design · JWT / Auth · SQL (MySQL, PostgreSQL)
Frontend	React.js · JavaScript (ES6+) · HTML5 · CSS3 · Responsive Design
DevOps & Tools	Git · GitHub · Docker · Linux / CLI · VS Code
Security	OWASP Top 10 · Web App Firewall · Cryptography APIs · Penetration Testing · CTF (Web Exploitation)

PROJECTS

AgroConnect DZ *React · Node.js/Express · PostgreSQL · REST API*

Full-stack agricultural marketplace for Algeria connecting 3 actor types (farmers, buyers, transporters). Architected a multi-role REST API with geolocation-based matching, an escrow payment module, and real-time GPS delivery tracking — handling all backend business logic and database schema design.

HoneyTrap — Multi-Protocol Honeypot System *Python · Flask · SQLite · Paramiko*

Cyber deception system emulating SSH, HTTP, FTP, and Telnet servers to capture attacker credentials and commands in real time. Built a custom SSH server with a stateful fake Linux filesystem (40+ commands, pipe chaining, output redirect) and an event-driven architecture with async GeoIP enrichment, per-thread SQLite connections, and a live attack map dashboard.

Exam Supervision Management System *Node.js · Express.js · MySQL*

Automated supervisor scheduling for a university faculty, eliminating manual assignment workflows. Engineered scheduling algorithms, a JWT-secured admin dashboard, and a full CRUD API with role-based access control.

Web Application Firewall *Python · Flask*

Modular security middleware with multi-layer threat detection (SQLi, XSS, SSRF). Implemented rate limiting, custom detection rules, and structured logging — reducing false-positive rates through pattern-tuned filtering.

Cryptography REST API *Python · Flask*

Backend API exposing AES, DES, RSA, RC4, SHA hashing, and digital signature endpoints. Designed clean, versioned REST endpoints with input validation and error standardization.

Games Store — Steam-like Platform *PHP · MySQL · JavaScript*

E-commerce platform with user auth, shopping cart, wishlist, and personal game library. Built a comprehensive admin dashboard with real-time sales analytics. Hardened against CSRF, XSS, and SQL injection using bcrypt and prepared statements.

Medical Cabinet Management System *Java · Swing*

Desktop application managing patients, doctors, appointments, and prescriptions. Applied structured OOP architecture with a clean separation of domain logic and UI layer.

EXPERIENCE

IT Internship · [Ministère du Commerce — Algérie](#) *Jul 2024*

- Assisted with network configuration and hardware maintenance in an enterprise IT environment.
- Diagnosed and resolved technical support issues, improving turnaround time for internal requests.

Web Exploitation Player · [CTF Competitions](#) *Jan 2026 – Present*

- Compete actively in CTF events targeting web vulnerabilities: SQLi, XSS, SSRF, and authentication bypasses.
- Translate offensive findings directly into defensive coding practices across personal and team projects.

Full-Stack Developer · [Hackathons](#) *Sep 2025*

- Delivered working full-stack demos under tight time constraints in team environments.
- Contributed to ideation, backend architecture, and final demo presentation.

EDUCATION

Engineering Degree in Computer Science — Cybersecurity · [USTHB, Algiers](#) *Sep 2023 – Present*

- 3rd year (expected graduation 2026). Core coursework: Web Security, Cryptography, Networks, Operating Systems, Databases, Algorithms.
- Active participant in technical clubs and hackathons.

Baccalauréat — Mathematics *Graduated with distinction — 16.08/20 (Très Bien)*

LANGUAGES

Arabic — Native **French** — Fluent **English** — Fluent (technical reading, writing, communication)